

Sorting through Affirmative Action: Three Field Experiments in Colombia

Marcela Ibanez, *Georg-August-Universität Göttingen*

Gerhard Riener, *Heinrich-Heine-Universität Düsseldorf*

The use of affirmative action policies to promote female employment remains debated. Do affirmative action policies attract female applicants, and does that come at the expense of deterring highly qualified male applicants? In three field experiments in Colombia, we compare job seekers who are informed of affirmative action selection criteria before they apply with those who are told after applying. We find that the gains in attracting female applicants far outweigh the losses in male applicants. Moreover, our results indicate that affirmative action does not decrease the quality of the top 15th percentile of the pool of applicants.

I. Introduction

Affirmative action policies continue to be a subject of intense debate (Cohen and Sterba 2003; Fullinwider 2011). Supporters point to their usefulness in addressing historical and statistical discrimination and to the advantages

We would like to thank Stephan Klasen and Ashok Rai for discussions and comments. We are grateful to Patricia Castro for her generosity and her invaluable help in conducting the experiments and to the Economics Department at the Universidad de los Andes, Bogotá, for their hospitality. This study was financed by the Courant Research Center for Poverty, Equity and Growth, University of Göttingen. Gerhard Riener gratefully acknowledges the financial support from the Deutsche Forschungsgemeinschaft under grant RTG 1411. Contact the corresponding author, Gerhard Riener, at gerhard.riener@gmail.com. Information concerning access to the data used in this paper is available as supplementary material online.

[*Journal of Labor Economics*, 2018, vol. 36, no. 2]

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Submitted April 15, 2015; Accepted April 3, 2017; Electronically published February 28, 2018

of diversity in both the workplace and the classroom (Clayton and Crosby 1992; Weber and Zulehner 2014). Critics argue that affirmative action is reverse discrimination (Newton 1973), violates the principle of merit (Walzer 1983:143–154), and can lead to economic inefficiencies (Coate and Loury 1993). In this paper, we study the sorting of job applicants in response to affirmative action in a naturally occurring labor market. We analyze some of the key questions in the debate: Do affirmative action policies for women actually encourage women to apply for jobs? Does this come at the cost of fewer applications from men? Do affirmative action policies foster less qualified women and deter the best-qualified men from applying? What kind of women are attracted to an affirmative action job? What kind of men are deterred?

We conducted the study in Colombia. Despite important improvements in gender disparities, gender inequality persists in the workplace (Peña et al. 2013; OECD 2014). Although the proportion of women who complete a university degree in Colombia is larger than that of men, there is a 23% gap in the labor force participation rate, the unemployment rate is 6.9% higher for women than for men, and on average, women earn 20% less and work 10 hours more per week than men (Ortiz et al. 2015). Similar figures are reported in other Latin American countries. In response to these inequalities, Colombia has adopted a comprehensive legal framework that promotes gender equity in the workplace (World Bank 2013). However, its implementation is rather weak. For example, the use of affirmative action policies has been applied mainly to the public sector, with varying degrees of compliance. In the private sector, affirmative action is not commonly used.

To investigate the effect of affirmative action on the labor market, we conducted three field experiments, which all use a similar design. We recruited a large pool of job seekers by posting job advertisements. Thereafter, interested job seekers were invited to apply for the job under two conditions of information. Half of the job seekers were told before they completed the application form that affirmative action would play a role in the selection process. The other half were informed of the affirmative action policy only after the application process was completed. This procedure allows us to measure the impact of affirmative action on the proportion of candidates of each gender who apply for the job advertised. Second, it allows us to observe the personal characteristics of job seekers before the policy is announced. Hence, we can investigate the sorting effects of candidates in terms of qualifications measured by their expected wages.

Affirmative action policies might reduce perceived discrimination in the hiring process for women and thereby induce more women to apply. However, this policy might eventually exacerbate the problem. For example, if women anticipate a patronizing equilibrium, in the spirit of Coate and Loury (1993), in which they are receiving the job as “token” females, they might choose not to apply, as they might perceive a reduction in their self-

image (Heilman et al. 1992; Unzueta et al. 2010; Bracha, Cohen, and Conell-Price 2013). Affirmative action could also deter male applicants and, in particular, result in a drop in applications from the best-qualified men.

Our main finding is that affirmative action encourages women to apply and that this does not come at the expense of a reduction in male applications. Furthermore, we find that the qualifications of male applicants are not systematically different under affirmative action, compared to the control group. Interestingly, our results suggest that in the absence of affirmative action, discrimination in the labor market, measured as gender differences in predicted wage, discourages male job seekers. Affirmative action is effective at closing this gap by attracting male job seekers who would otherwise be discouraged from applying.

To the best of our knowledge, this is the first paper that provides field evidence on the impact of affirmative action policies in favor of women on sorting in the labor market. Field experiments have been used before to examine the sorting of workers into different jobs due to dimensions different from the one we consider here.¹

Laboratory experiments that consider self-selection into a tournament support the use of affirmative action rules to incentivize women to enter into competitive environments (Balafoutas and Sutter 2012; Calsamiglia, Franke, and Rey-Biel 2013; Niederle et al. 2013; Balafoutas, Davis, and Sutter 2016). Our field experiment allows us to capture dimensions different from aversion to competition, which could constrain women from participating in the labor market. For example, cultural norms toward female employment, the availability of childcare, and support at home can play an important role in explaining female labor market participation (Coen-Pirani et al. 2010; Barone and Mocetti 2011; Fogli and Veldkamp 2011; Fernández 2013; Bauernschuster and Schlotter 2015). Moreover, unlike lab experiments, participants in our field experiment were unaware of the fact that they were participating in an experiment. Therefore, our results are not subject to experimenter-demand effects.²

The remainder of the paper is organized as follows: Section II presents the conceptual framework. Section III describes the local context of the labor

¹ For instance, Bonin et al. (2007) consider the effect of wage volatility; Kluge and Schaffner (2007) consider health risk; Guiteras and Jack (2014) and Dal Bó et al. (2013) consider the effect of value of the compensation; Ashraf, Bandiera, and Kelsey (2014) consider the salience of career incentives; Dohmen and Falk (2011) and Flory et al. (2015) consider the effect of competitive versus individual remuneration schemes; Barbulescu and Bidwell (2013) consider the type of job (male vs. female stereotypical jobs); and Lefebvre and Merrigan (2008) and Havnes and Mogstad (2011) consider the impact of provision of child care.

² We use the same terminology and definitions as Al-Ubaydli and List (2012) and acknowledge the “leap of faith” required in making this statement. This is, however, common in all types of empirical research.

market in Colombia, while Section IV presents the experimental design and procedures. The results are presented in Section V. Finally, we conclude with a discussion in Section VI.

II. Conceptual Framework

Using a simple theoretical framework similar to the one used by Ashraf et al. (2014), we consider how the salience of affirmative action that gives preference to women over men within the hiring process affects the propensity of male and female job seekers to submit an application. In addition, we consider the impact that it has on the qualifications of those job seekers.

We consider a labor market with gender discrimination that, on average, gives advantages to men over women, both on the likelihood of being employed and on the offered wage. We assume that in the absence of affirmative action rules, the individual cannot distinguish between discriminatory and nondiscriminatory firms. Affirmative action makes salient the use of reverse discrimination during the recruitment process, changes job seekers' perception of the probability of being employed, and thus changes their respective incentives to submit an application. We focus exclusively on the sorting effects on the pool of job seekers and do not consider general equilibrium effects.

Labor market context. We consider risk-neutral individuals who differ in terms of gender, $s \in \{m, f\}$ and qualifications, $q \in [0, 1]$. Individuals with higher qualifications are more likely to be employed and receive a higher salary, so $\partial w^k / \partial q > 0$ and $\partial \pi^k / \partial q > 0$. However, individuals confront a discriminatory labor market that—on average—favors male candidates. For a given level of qualifications, male agents on average expect higher likelihood of being employed than female agents— $\pi_f(q) < \pi_m(q)$ —and on average receive a higher salary— $w_f(q) < w_m(q)$.

A firm n posts a job offer, $o = \{w^n, G\}$, that is characterized by a salary w^n and the salience of affirmative action, $G = \{NAA, AA\}$, where NAA denotes no use of affirmative action and AA the use of affirmative action. We do not model the decision on the levels of w and G within the firm and take them as exogenous. The salary w^n corresponds to the average salary that a candidate with qualifications $\bar{q}$ would receive. Firm n is nondiscriminatory. Therefore, the salary that it offers is the same for men and women. Yet, compared with other firms, firm n offers a relatively higher salary for women and a lower salary for men, $w_f^k(\bar{q}) < w_f^n(\bar{q}) = w_m^n(\bar{q}) < w_m^k(\bar{q})$. The agent does not have other information on how discriminatory firm n is before applying. Besides, the wage offer alone does not allow the job seeker to identify whether or not the firm is discriminatory. If affirmative action is not salient ($G = NAA$), individuals would expect discrimination in favor of male job seekers, $\pi_f^{NAA}(q) < \pi_m^{NAA}(q)$.

An individual i decides whether to submit an application for an offer with firm n , given that he or she has an opportunity cost, $v_s(q)$. The opportunity cost of applying depends on the likelihood of being employed at another job k , times the expected salary at that job: $v_s(q) = \pi_s^k(q)w_s^k(q)$. The actual cost of completing an application b is minimal, so it can be disregarded.

When the job offer, n , does not make salient the use of affirmative action rules, applicants submit an application if the expected wage in firm n is larger than the expected opportunity cost:

$$\pi_s^{\text{NAA}}(q)w^n - \pi_s^k(q)w_s^k(q) > 0. \quad (1)$$

This leads to our first proposition:

PROPOSITION 1: In a discriminatory market, where firms set a nondiscriminatory wage and do not signal their employment prospects for women, the best-qualified female candidate will be better than the best-qualified male applicant.

Proof: When firms do not signal their type, $\pi_s^{\text{NAA}} = \pi_s^k$. This implies that agents would apply for the position only if $w^n > w_s^k(q)$. However, since w^n corresponds to the average wage for the candidate with qualifications $\bar{q}$, this condition implies that agents with higher level of qualifications, $q > \bar{q}$, whose expected salary is above the offered salary, do not apply for the job offer n . A nondiscriminatory firm would set salaries so $w_i^k(\bar{q}) < w_i^n(\bar{q}) = w_m^n(\bar{q}) < w_m^k(\bar{q})$. The best-qualified men, who face high opportunity cost, would not accept the job offer and thus drop from the application process. On the other hand, as $w_i^n > w_i^k(\bar{q})$, women with qualifications above $\bar{q}$ would be attracted to the job offer.

If the firm makes affirmative action salient ($G = \text{AA}$), we assume that individuals consider this as an informative signal of the conditions during the recruitment process. Therefore, female agents perceive $\pi_i^{\text{NAA}}(q) < \pi_i^{\text{AA}}(q)$, and male agents perceive $\pi_m^{\text{AA}}(q) < \pi_m^{\text{NAA}}(q)$.

PROPOSITION 2: Making affirmative action policies that favor women salient leads to (1) an increase (decrease) in the proportion of female (male) agents who apply, compared with the firm that does not use affirmative action. Moreover, it leads to (2) an increase (decrease) in the overall level of qualifications among the female (male) subject pool.

Proof: Making affirmative action salient implies that $\pi_m^{\text{AA}}(q) < \pi_m^k(q)$ and $\pi_i^{\text{AA}}(q) > \pi_i^k(q)$. The expected earnings are therefore higher for women when affirmative action is used than when it is not used, thus increasing

the application rate by female agents and having the opposite effect on male candidates. Equation (1) implies that a candidate applies if

$$\frac{w_s^k(q)}{w} \leq \frac{\pi_s^{AA}(q)}{\pi_f^k(q)}.$$

Under affirmative action, $w_m(q)/w < \pi_m^{AA}(q)/\pi_m^k(q) < 1$ and $w_f(q)/w \leq \pi_f^{AA}(q)/\pi_f^k(q) > 1$. That is, male agents who expect to receive a salary below the offered salary $w_m^k(q) < w_m^k(\bar{q}) = w$ (those who have a below-average level of qualifications) and women who expect to receive a salary above the offered salary $w_m^k(q) > w_m^k(\bar{q}) = w$ (those who have an above-average level of qualifications) apply for job offer n .

III. Context

To test the predictions of the above model and evaluate the sorting effects of affirmative action rules, we conducted three labor market field experiments in Colombia. The labor market in Colombia—as in other countries in Latin America—offers conditions that are less favorable for women than for men (UNDP 2013). Although less pronounced, gender disparities are also observed for the proportion of the population holding a university degree. Cepeda and Baron (2012) show that only 42% of the gender gap in wages for recently graduated students can be explained by selection into different areas of study. Table 1 presents descriptive statistics of the national representative survey of recently graduated students conducted by the Colombian Ministry of Education in 2010. Panel A presents the descriptive statistics for graduates with a bachelor's degree, while panel B refers to graduates with a master's degree. While women represent a larger share of recent graduates, they face worse employment conditions. In four of the eight areas of study, the employment rate of women is significantly lower than that of men. Inequalities in the labor market are also observed in average income. Depending on the area of study, the average earnings of females is between 9% and 20% lower than those of male graduates with a bachelor's degree.

IV. Experimental Procedures and Design

Three experiments form the basis of our study. In the first two experiments, which we refer to as the “Assistant 1” (hereafter A1) and “Assistant 2” (hereafter A2) experiments, we recruited applicants for research-assistant positions for projects of the two coauthors of this paper. Candidates were required to have completed or be close to completing a bachelor's degree in any area of study. No previous work experience was requested. The third experiment was conducted in collaboration with a private consultancy com-

pany that offered one position for a consultant with at least 2 years of professional experience in implementation and evaluation of community development projects. We refer to this experiment as the “Consultant” (hereafter C) experiment. Interested candidates were required to have at least a bachelor’s degree in a relevant field of study that included public administration, public management, or economics. The recruitment strategy in the three experiments used a design similar to that in Flory et al. (2015) and involves five stages that are presented in table 2 and described in more detail below.

A. Stage 1: Announcement

In the first stage, we announced the positions through newspapers, university employment boards, social media, and e-mail lists. In this announcement we provided general information about the position. Table 3 presents a summary of the information that participants received in this and subsequent stages.³ In this stage we tried to get a large pool of subjects interested in the positions over which we could randomize the treatments. A similar procedure was used by Flory et al. (2015) in examining the willingness to enter into competitive environments.

For the A1 and A2 experiments, we invited applicants from all areas of study, to compare how discrimination affects applications. For the C experiment, the position was also announced via a “hot” mailing list containing around 3,000 e-mail addresses of currently active consultants. Because of the demands of the company, candidates were required to have at least 2 years of experience.

B. Stage 2: Statement of Interest

Job seekers were allowed to state their interest in the position by filling out a low-hurdle statement-of-interest form.⁴ The announcement elicited great interest, and in the three experiments about 5,000 people expressed interest in the positions.

In this stage, job seekers provided information about their gender, age, level and area of studies, and year of graduation.⁵ The A2 experiment also included additional questions on the individual qualifications and personalities from the pool of applicants. Among the qualification measures, we included grades during studies, the Frederick (2005) cognitive-reflection test (CRT), and an exercise on a work example of digitizing data (see Sec. A4 of the appendix). Personality questions were assessed with the Spanish version of the Big 5 personality test (Benet-Martinez and John 1998). Finally, we

³ Section A1 of the appendix presents the announcements used.

⁴ Table A1 presents the complete list of socioeconomic information collected in this stage

⁵ Asking for gender is common in Colombia, and there are no legal implications in doing so.

Table 1
The Labor Market in Colombia

	Graduates		Participation Rate		Employment Rate		Monthly Income (000 COP)		Standard Deviation of Income		
	Proportion	Female	Male	Female	Male	Female	Male	Female	Male	t-test	
											PrTest
A. Bachelor's											
Agricultural sciences	.016	.342	.980	.962	.805	.735	1,502.618	1,624.447	1,183.221	1,569.556	
Fine arts	.042	.569	.980	.938	.816	.786	2,011.802	2,046.965	2,964.940	3,286.117	
Education sciences	.062	.592	.983	.965	.772	.779	1,396.038	1,328.939	1,235.201	1,576.196	
Health sciences	.078	.783	.903	.961	.836	.768	2,243.73	1,791.756	1,865.875	1,816.761	
Social sciences	.200	.747	.982	.971	.814	.764	2,289.501	2,131.564	2,527.741	2,577.776	
Economics, business	.243	.605	.986	.971	.860	.841	2,380.529	2,080.73	3,120.038	2,736.778	
Engineering	.350	.411	.974	.971	.844	.816	2,201.829	1,990.734	2,419.758	1,868.403	**
Natural and sciences	.027	.576	.909	.919	.704	.685	2,117.082	1,837.311	2,370.282	1,909.170	**
Total	1.000	.576	.974	.967	.834	.796	2,177.805	1,981.455	2,488.694	2,307.781	**
Observations	18,822	18,822	7,986	10,836	7,779	10,475	6,286	8,102	6,286	8,102	

B. Master's												
Agricultural sciences	.002	.500	1.000	1.000	.750	.750	1,833.333	1,363.636	577.3503	236.1887		
Fine arts	.007	.520	.917	.923	1.000	1.000	2,793.388	2,511.995	1,147.646	1,641.646		
Education sciences	.062	.500	1.000	.972	*	.981	.895	2,111.026	2,142.841	1,532.941	2,188.622	
Health sciences	.035	.637	.956	1.000	*	1.000	.886	*	4,859.885	2,894.904	**	6,826.066
Social sciences	.160	.593	.978	.964		.888	.882	2,805.042	2,622.217	2,363.556	2,895.943	
Economics, business	.550	.522	.987	.972	**	.925	.884	3,526.48	3,019.666	3,720.314	3,764.161	
Engineering	.170	.363	.961	.982		.942	.915	2,956.989	3,078.764	1,899.327	2,869.561	
Natural and sciences	.013	.511	.913	.958		1.000	.913	2,421.515	3,528.499	1,554.137	4,547.97	
Total	1.000	.509	.979	.973		.931	.890	3,227.139	2,893.423	3,027.854	3,343.056	
Observations	3,505	3,517	1,722	1,783		1,685	1,734	1,516	1,498	1,516	1,498	

NOTE.—This table reports labor market statistics constructed by the authors on the basis of the 2010 survey of recently graduated university students (RGUS; Encuesta de Seguimiento a Graduados) from the Ministry of Education. The statistics are calculated for students who graduated between 2001 and 2007. Monthly income is standardized, considering the number of hours worked over the past week and projected for 40 hours of work per week. *Proportion test^a (PrTest) refers to the test of the null hypothesis of equal proportions of male and female graduates participating in the labor market or the proportion of male and female participants in the labor market employed. The Student *t*-test tests the hypothesis of equal means in the standardized monthly income by male and female graduates who were employed at the moment of the survey. COP = Colombian pesos.

* $p < .1$.
 ** $p < .05$.
 *** $p < .01$.

Table 2
Recruitment Process

Stage	Assistant 1 (A1)			Assistant 2 (A2)			Consultant (C)		
	Dates	Number	% Female	Dates	Number	% Female	Date	Number	% Female
1. Announcement	Oct. 10–Dec. 10	2,207	55.14	Jul. 13–Aug. 13	2,341	49.17	Jan. 11	310	43.87
2. Statement of interest	Oct. 10–Dec. 10	733	55.53	Jul. 13–Aug. 13	761	50.46	Jan. 11	293	46.42
3. Randomization	Dec. 10	311	54.05	Aug. 13	367	47.96	Jan. 11	91	41.76
4. Application	Dec. 10–Jan. 11	3	100.00	Aug. 13	22	50.00	Jan. 11	1	100.00
5. Hiring	Feb. 11			Sept. 13			Feb. 11		

Table 3
Information Provided to Participants

	Assistant 1 (A1)	Assistant 2 (A2)	Consultant (C)
Stage 1: announcement:			
Employer	University	University	Firm
Location of job	Rural	Urban	Rural
Position	Research assistant	Research assistant	Consultant
Responsibilities	Data collection	Data entry	Training
Stage 4: invitation to apply:			
Salary (per month)	1.5 m COP	Performance based	3 m COP
Duration	2 months	1–2 months	4 months

NOTE.—COP = Colombian pesos.

included questions from the German Socioeconomic Panel on risk aversion. The question on risk aversion has been validated in a study by Dohmen et al. (2011) and was shown to have high correlation with different measures of risk aversion for distinct domains of risk.

The measures that we obtained in this stage can be considered exogenous, as we observed them before participants were exposed to the treatment. These measures are used to evaluate the qualifications of job seekers, as explained in more detail below. With self-reported measures, truthful reporting is of some concern, as people have a tendency to misreport in order to increase their chances of being hired. We therefore provided incentives for truthful reporting by announcing that supporting documents would be required in order to be eligible for employment if selected. This policy was communicated whenever job seekers had to enter verifiable information during the application process.

C. Stage 3: Invitation to Apply

In this stage, about half of the job seekers who submitted a statement-of-interest form were randomly assigned to be invited to continue the application process under two conditions: the AA treatment or a control group. This group, which we refer to as “invited job seekers,” received an e-mail explaining the particular conditions of employment regarding job responsibilities, salary, and duration of the employment and an invitation to apply to the job (see table 3).⁶

In the AA treatment, the last line of the invitation letter announced that the employer is an equal-opportunity employer and that during the hiring process women would be favored. Invited job seekers under the AA treatment were exposed to the affirmative action statement before starting the extensive application process. In the control group, we presented invited job seekers with the affirmative action statement only after they had fin-

⁶ Section A3 of the appendix presents the invitation letters used in the experiments.

ished the questionnaire. Therefore, we achieve *ex post* equality of information for subjects who completed the questionnaire and effectively applied to the job.

We used two different affirmative action rules that are commonly used to favor female applicants. The A1 and A2 experiments used a quota rule in which a fixed share of the positions are reserved for women. The following statement (translated from Spanish) was displayed to invited job seekers in the AA treatment:

The University of . . . is an equal-opportunities employer. To increase female participation in areas where women are up to now underrepresented, a minimum of 50% of the hired assistants will be women.

The C experiment used a preferential-treatment rule and presented the following statement:

We are an equal-opportunity employer who seeks to increase the participation of women in areas where they have been underrepresented. For equally qualified candidates, women will be preferred.⁷

Finally, we asked invited job seekers to complete a lengthy application questionnaire in order to apply for the position.⁸ Filling out the application questionnaire took between 40 and 60 minutes and required a search for supporting information on several questions, which was announced to job seekers. By using a demanding and time-consuming application questionnaire that would increase the cost of the application (time required), we wanted to capture the effect of the opportunity cost of applying to the position. As recruitment processes are usually very comprehensive, we expected that invited job seekers would not be surprised about the demanding application process.⁹

D. Stage 4: Application

Invited job seekers had access to a personalized page and could complete the application form over different sessions, saving the information and

⁷ The original statement in Spanish is presented in the supplementary material online.

⁸ The A1 and A2 experiments are similar in terms of implementation and especially the affirmative action policy, while the C experiment, although similar in spirit, deviates in the affirmative action policy applied and the decision to apply it, which was done not via e-mail but via the home page itself.

⁹ In the A1 experiment, we stratified job seekers into AA treatment and control groups on the basis of used gender and the main residence in Bogotá, while in the A2 experiment, we also added level of studies (undergraduate or master) in the stratification. In the C experiment, randomization was done immediately after the statement of interest was completed and used a random-number generator within the survey software (LimeSurvey 2012). Job seekers had a 50% probability of being selected into the AA treatment group. Since allocation was truly random, 45% of the job seekers ended up being allocated into the AA treatment.

continuing the application over several days. Yet a strict deadline was set after which no application was accepted. The measures of qualifications obtained in this stage are endogenous, as performance on the tests and responses to the questionnaire could have been affected by the treatment. In this paper, we are interested in the sorting effects of affirmative action—hence, in the analysis we do not consider the effects that the policy might have had on the responses given in the application form.

Invited job seekers who were not interested in the job also had the chance to actively drop out of the application process by either clicking a “disagree” button (A1 and A2 experiments) or completing a short exit questionnaire (C experiment). In the questionnaire, we asked the reasons why they left the application process, especially whether they disliked the affirmative action policy applied to those who were randomized into the AA treatment group. However, the turnout was low, and only 3.2% of subjects who did not start the application process actively dropped out.

Field experiments that go over multiple sessions and are not conducted at the same time could suffer from information spillovers. We tried to minimize this by opening the positions at the same time and by recording the starting time of the applications, in order to control for potential timing effects.

E. Stage 5: Hiring

Invited job seekers who completed the application processes, whom we refer to as “applicants,” were ranked on qualifications. The top 10 applicants were invited for an interview. The best applicants received a job offer. In the A1 experiment, three people were hired (all of them women).¹⁰ Two of them were hired for 2 months, and a third person was hired for 4 months. We hired 22 applicants in the A2 experiment, half of them female. In the C experiment, one female applicant was hired for 6 months. In this paper, we focus on the analysis of the sorting effects in application process. We do not consider measures of on-the-job performance, as the limited number of positions offered does not allow us to conduct a statistical analysis of the impact of affirmative action on job performance.

V. Results

A. Descriptive Statistics and Randomization Checks

The number of participants and the proportion of female participants in each of the stages of the experiment are presented in table 2. The response

¹⁰ Affirmative action was intended to favor women; hence, we did not intend to have equal gender representation. In other words, it should not have been expected that at least 50% of the positions would be reserved for men.

rate to the invitation to apply to the position was high, and 43% of the invited candidates submitted the complete application form. About half of the submitted applications were from women.

1. Representativeness of the Job Seekers

One question that arises, given our experimental design, is whether job seekers who stated interest in the position are representative of the total population of job seekers. Two considerations are worth noting. First, the announcements for the A1 and A2 experiments were directed to recently graduated students. However, job seekers from particular fields of study or a particular gender could have self-selected into the experiment if they considered themselves better qualified for the position. Hence, we did not expect to have a representative sample of all job seekers or recent graduates. Second, our intention in this experiment was to evaluate how affirmative action rules affect sorting in the labor market. Therefore, random allocation of job seekers into the treatment and control groups enables us to evaluate the behavioral response for a comparable population group. Since we know the total number of job seekers, we can measure the impact of affirmative action as the difference in the proportion of participants of a given gender who apply in the treatment group, compared with that in the control group. Therefore, this procedure is preferable to an alternative design that relies on sending different announcements.

To test how representative our sample is, in Section A6 of the appendix we compare the socioeconomic characteristics of the job seekers in our experiment with those in the survey from the Colombian Ministry of Education of recently graduated university students (RGUS) in 2010. Not surprisingly, we find that our sample differs in many aspects from the sample of recently graduated students. This can be explained by the characteristics of the jobs we posted.

2. Randomization Tests

Another important question is whether the invited job seekers assigned to the different treatments are comparable in observable characteristics. Table 4 shows the descriptive statistics of the invited job seekers in the different treatments and experiments. The pool of invited job seekers in the A1 and A2 experiments is, on average, 28 and 24 years old, respectively. This group is relatively less educated and less experienced than the invited job seekers in the C experiment. Comparing the subjects' characteristics after randomization with a joint orthogonality test, we find that most of the observable characteristics of the invited job seekers have good balance across treatments for each gender. Yet we find a slight unbalance in performance in the CRT for female applicants in the A2 experiment (t -test: $p < .10$). In the analysis we control for this difference.

B. Impact of Affirmative Action in Sorting by Gender

One advantage of our experimental design is that we can compare the proportions of invited job seekers that submit the application form—the application rate—under both conditions. Application rates are 42.2%, 48.3%, and 29.7% in the A1, A2, and C experiments, respectively. Overall, we find that the application rates (not disaggregating by gender) are not significantly different under the control and AA treatments for any of the experiments (Fisher exact tests: $p = .709$ for A1; $p = .718$ for A2; and $p = .533$ for C).

The more interesting question is how the gender composition of applicants changes under affirmative action. Figure 1 shows the raw application rates for men and women under the different treatments and experiments. We find that in the control treatment, women are significantly less likely to apply than men in two of the experiments, the A2 and C experiments (Fisher exact test: $p < .01$). Under the AA treatment, the gender difference in application probabilities completely vanishes, with women being equally likely to apply as men in all three experiments (Fisher exact test: $p > .10$).

RESULT 1: The affirmative action statements eliminate the gender gap in the application rates in all three experiments.

Figure 1 suggests that the shrinking of the gender gap in application rates is due not only to a higher application rate by women but also to a lower application rate by men. We find that this is not the case and that the application rate for men under the AA treatment is not significantly different from that in the control treatment in any of the experiments (Fisher exact test, $p > .10$).

To further examine this result, we estimate the following linear probability model:

$$\text{completed}_i = \alpha + \beta_1 \text{AA}_i + \beta_2 \text{female}_i + \beta_3 \text{AA}_i \times \text{female}_i + \varepsilon_i. \quad (2)$$

The dependent variable “completed” is binary and takes a value of 1 for job seekers who submitted a completed application form and 0 otherwise. We focus on completed applications, as this is the economically relevant variable from the point of view of the employer.¹¹ The variable “AA” refers to the treatment variable and takes a value equal to 1 for job seekers randomly selected into the AA treatment and 0 otherwise. The variable “female” is a dummy variable that indicates whether the job seeker is a woman.

¹¹ Alternatively, one could also be interested in the affective reaction to the affirmative action statement by considering whether or not people started the application process. We find that all results hold true when looking also at subjects who agreed to the conditions and started the application procedure but did not necessarily finish it.

Table 4
Randomization Checks

	Control			Affirmative Action (AA) Treatment			<i>p</i> -Values by AA			
	Overall	Female	Male	Overall	Female	Male	Overall	Female	Male	
	A. Assistant 1 (A1)									
Experience (years)	3.227 (.244)	2.864 (.266)	3.673 (.432)	3.325 (.238)	3.080 (.272)	3.627 (.412)	3.276 (.170)	.774	.570	.939
Master's (proportion)	.104 (.016)	.098 (.021)	.110 (.025)	.093 (.015)	.094 (.020)	.092 (.023)	.098 (.011)	.629	.879	.583
Age (years)	27.512 (.342)	27.029 (.405)	28.117 (.577)	27.861 (.321)	27.084 (.375)	28.828 (.542)	27.686 (.235)	.458	.922	.369
Bogotá (proportion)	.362 (.025)	.382 (.034)	.337 (.037)	.363 (.025)	.384 (.034)	.337 (.037)	.363 (.018)	.978	.969	1.000
Observations	367	204	163	366	203	163	733			
B. Assistant 2 (A2)										
Experience (years)	3.600 (.213)	3.592 (.250)	3.609 (.347)	3.509 (.179)	3.410 (.264)	3.609 (.242)	3.554 (.139)	.743	.618	1.000
Master's (proportion)	.040 (.010)	.042 (.015)	.037 (.014)	.049 (.011)	.052 (.016)	.047 (.015)	.045 (.007)	.518	.662	.633
Age (years)	24.846 (.339)	24.571 (.472)	25.123 (.486)	24.660 (.312)	24.415 (.443)	24.910 (.440)	24.752 (.230)	.686	.809	.746
Risk-taking person	5.772 (.117)	5.611 (.168)	5.936 (.164)	5.747 (.117)	5.624 (.164)	5.874 (.169)	5.760 (.083)	.883	.955	.792
Patient person	3.395 (.108)	3.447 (.159)	3.342 (.145)	3.214 (.101)	3.392 (.144)	3.032 (.141)	3.304 (.074)	.219	.796	.125

Impulsive person	8.109 (.104)	8.189 (.137)	8.027 (.157)	7.948 (.109)	7.892 (.155)	8.005 (.154)	8.028 (.075)	.286	.151	.922
Relative grade (avg./max.)	.903 (.034)	.901 (.040)	.905 (.056)	.893 (.029)	.859 (.011)	.928 (.057)	.898 (.022)	.833	.310	.767
CRT score	1.369 (.058)	1.232 (.082)	1.508 (.080)	1.242 (.060)	1.000 (.081)	1.489 (.084)	1.305 (.042)	.128	.045**	.873
Score: work example	18.971 (.140)	19.279 (.157)	18.759 (.209)	19.065 (.120)	19.263 (.132)	19.063 (.143)	19.018 (.092)	.610	.938	.231
Has children	.130 (.017)	.142 (.025)	.118 (.024)	.135 (.017)	.144 (.025)	.126 (.024)	.133 (.012)	.825	.951	.798
Number of children	.175 (.026)	.174 (.034)	.176 (.040)	.193 (.029)	.201 (.042)	.184 (.039)	.184 (.019)	.648	.611	.890
Observations	378	190	187	386	194	190	764			
C. Consultant (C)										
Experience (years)	8.904 (.584)	10.937 (1.009)	7.300 (.631)	9.377 (.556)	10.354 (.905)	8.428 (.640)	9.117 (.406)	.558	.668	.211
Master's (proportion)	.345 (.037)	.282 (.054)	.422 (.052)	.324 (.039)	.308 (.058)	.388 (.060)	.335 (.027)	.693	.742	.668
Observations	168	71	90	142	65	67	310			

NOTE.—This table reports the results of the randomization check. Standard deviations are reported in parentheses. In the Consultant (C) experiment, 17 subjects did not report their gender and experience, and one person did not report the university of study. The last set of columns reports the *p*-value from a joint orthogonality heteroskedasticity robust test of treatment allocation. Conducting 54 tests and not controlling for multiple testing, we observe only one significant difference at the 5% level, which is less than expected at this level. This indicates that the randomization procedure was successful. CRT = cognitive-reflection test score.

** Significant at the 5% level.

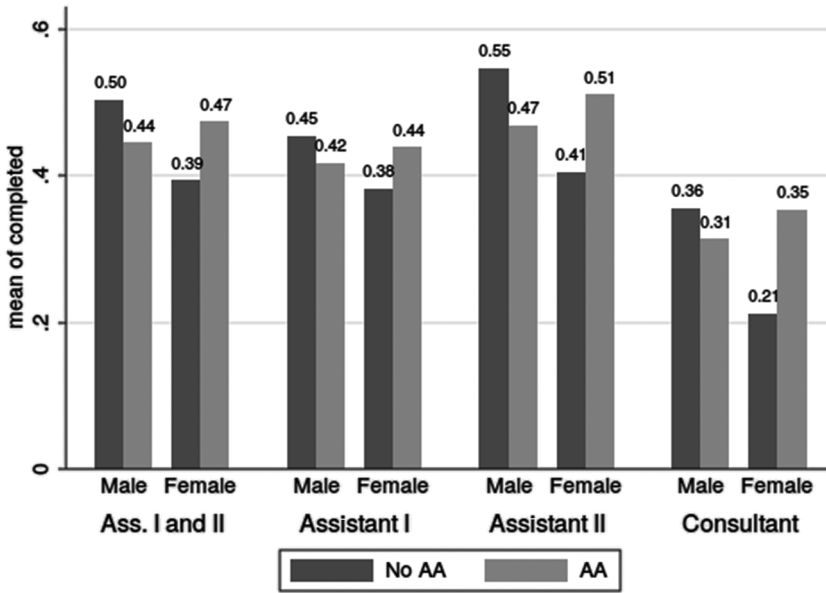


FIG. 1.—Share of completed applications by study and gender. “Ass. I and II” pools the data from the two assistant experiments (A1 and A2). Fisher exact tests indicate that affirmative action (AA) increases the share of women applying in the pooled assistant experiments as well as in the A2 and the consultant (C) experiments separately (Fisher exact test: $p = .026, .041, \text{ and } .085$, respectively), and we do not observe a significant reduction for men.

The estimation results are presented in table 5.¹² We report the results by experiment and the pooled results for all experiments. Panel A presents the results of the regression model, while panel B presents the contrasts of the effect of affirmative action and gender differences by treatment. For each model, we present the results with and without controls on socioeconomic characteristics.

We find that between 43% and 57% of the male job seekers completed the application process. As indicated by the contrasts, in the absence of affirmative action, women were significantly less likely to apply to the positions than men, with a difference between 7 and 24 percentage points, depending on the experiment. The results also indicate that affirmative action closes the gender gap in applications: the difference in application rates between men and women, as given by $\beta_2 + \beta_3$, is not significantly different from 0 in any of the three experiments.

From a policy perspective it is important to understand whether the gap is closed by having more women applying, as is expected, or whether this ef-

¹² Our results are robust to nonlinear estimations such as probit and logit models.

fect is due to fewer men applying. The contrast results in table 5 indicate that in all three experiments women are significantly more likely to apply in the AA treatment than in the control treatment. The application rate is 5–22 percentage points higher for women under the AA treatment than under the control treatment. Application rates for men are not significantly different in the AA treatments and the control treatment in the pooled regressions and in two of the three experiments. However, in the A2 experiment, men are 6–8 percentage points less likely to apply than in the control treatment. That shortfall is compensated for by an increase in female applicants in that experiment.¹³

RESULT 2: Affirmative action policies induce significantly more women to apply in all experiments. There is a significantly negative effect on application rates for men in experiment A2 under the AA treatment.

To check the robustness of the results, we examined the timing of the applications. If women in the control treatment applied later, this would indicate the presence of spillovers between the AA treatment and control groups. We do not find any significant differences in the timing of the applications (neither statistically nor in size).

C. Affirmative Action and Qualifications of the Applicant Pool

While affirmative action seems to encourage more women to apply, it is not clear whether this comes at a cost of attracting candidates who are less qualified. In this section, we explore how costly affirmative action policies are in terms of potential changes in the composition of characteristics of the applicant pool.

It is reasonable to expect that more-qualified job seekers are more likely to be employed and receive a higher salary. Hence, we aggregate different measures of qualifications that we elicited in the statement of interest form (i.e., experience, area, and level of studies) in one measure: predicted wage.

Predicted wage is constructed as follows. First, using secondary information, we estimate a model based on Gronau's (1974) model of wage offers (y as defined by eq. [3]) and employment (e as defined by eq. [4]):

$$y = \alpha^w + \beta^w Z^w + u, \quad (3)$$

$$e = 1(\alpha + \beta^u Z_i^u + v > 0), \quad (4)$$

¹³ As the coefficients on affirmative action are large for the A1 and C experiments (−0.043 and −0.086, respectively), we conducted power tests to determine the sample size that would be necessary to detect effects of this empirically measured size at a significance level of 5% and a power of 80%, using χ^2 tests. For the A1 experiment, this would require 2,848 observations per treatment and for the C experiment 1,969 per treatment.

Table 5
Linear Probability Model of Completed Applications under Affirmative Action

	Pooled		Assistant 1 (A1)		Assistant 2 (A2)		Consultant (C)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	
A. Regression								
Affirmative action (AA)	-.049 (.034)	-.037 (.029)	-.035 (.027)	-.087*** (.029)	-.086*** (.031)	-.042 (.076)	-.034 (.072)	
Female	-.107*** (.033)	-.072* (.041)	-.077** (.037)	-.133*** (.028)	-.136*** (.026)	-.144** (.070)	-.108 (.069)	
AA × female	.140*** (.047)	.093** (.039)	.091** (.038)	.179*** (.057)	.174*** (.055)	.185* (.109)	.169* (.103)	
Experience			-.004 (.009)		.015** (.007)		.003 (.004)	
Master			.029 (.075)	-.149** (.063)			.331*** (.057)	
Age			.001 (.005)		-.003 (.004)			
Has children					-.130*** (.047)			
CRT					-.002 (.015)			
Score: work example					.010 (.006)			
Baseline	.473*** (.024)	.454*** (.043)	.449*** (.131)	.571*** (.045)	.428*** (.149)	.356*** (.051)	.196*** (.057)	
Observations	1,787	733	721	694	689	293	293	

		B. Contrasts					
AA vs. no AA:							
Male		-.049	-.037	-.035	-.087	-.086	-.042
		(.034)	(.029)	(.027)	(.029)	(.031)	(.076)
Female		[.149]	[.208]	[.198]	[.004]	[.006]	[.582]
		.091	.056	.057	.092	.088	.143
Female vs. male:		(.032)	(.027)	(.027)	(.040)	(.034)	(.077)
		[.005]	[.041]	[.041]	[.023]	[.012]	[.065]
No AA		-.107	-.072	-.077	-.133	-.136	-.144
		(.033)	(.041)	(.037)	(.028)	(.026)	(.070)
AA		[.001]	[.084]	[.041]	[.000]	[.000]	[.041]
		.033	.021	.014	.046	.038	.040
		(.033)	(.034)	(.036)	(.042)	(.044)	(.083)
		[.326]	[.536]	[.700]	[.273]	[.391]	[.625]
							[.422]

NOTE.—Panel A reports the results of a linear probability model on completed application for all data pooled (col. 1) and the A1 (cols. 2, 3), A2 (cols. 4, 5), and C (cols. 6, 7) experiments separately. Panel B reports the differences in application rates by treatment and by gender resulting from panel A. Standard errors are reported in parentheses and p -values in brackets. “AA” indicates the affirmative action treatment employed for the experiment: a quota rule for A1 and A2 and a preferential-treatment rule for C. Standard errors are clustered at the city level for the A1 experiment and at the university of graduation level for A2. The C experiment and the pooled models report heteroskedasticity robust standard errors. The discrepancy between the total number of observations and the sum of the reported observations is that not all subjects reported the city or the university of graduation, so these persons have been omitted in the individual regression. Significance levels are for results of t -test. CRT = cognitive reflection test score.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

where α^i, β^i , for $i \in \{w, u\}$, are column vectors of coefficients; Z^w and Z^u are column vectors of the characteristics of the job seeker and its interaction terms with gender; and u and v are random variables. The elements of Z^w are a strict subset of the elements of Z^u , and the elements and exclusion restrictions vary by experiment because of data availability, as described below. To estimate this model, we employ a Heckman (1979) sample selection model. In the first stage, we estimate the likelihood of being employed by using a probit model. In the second step, we estimate an ordinary least squares regression model for wages that includes the inverse Mills ratio to account for self-selection in employment. A quadratic term of the applicant's age and experience in Z relaxes the assumption of a linear relationship that may lead to an underrejection of the null hypothesis with no effect of the variables of interest, as has been shown by Løken, Mogstad, and Wiswall (2012). We use a parametric approach, adding a quadratic term to the linear formulation, as we have variables with far more categories and also continuous independent variables. Section A7 of the appendix presents the estimated coefficients and discusses the findings.

As exclusion restrictions we use experience squared and its interaction with gender in the A1 and C experiments. In addition, the A2 experiment includes the number of children and its interaction with gender. This wage equation explains how the labor market rewards productive attributes such as experience, area of study, and level of studies. We use data from the World Bank (2014) STEP (Skills Toward Employability and Productivity) survey, which is a representative-sample survey of the Colombian working population on cognitive and noncognitive skills. The main advantage of this survey is that it includes most of the variables for qualifications and personality that we elicited before treatment allocation.¹⁴

Second, we use the estimated parameters from equation (3) to predict expected wage, $\hat{w} = \hat{\epsilon} \times \hat{y}$ —setting the predicted wage for the unemployed to 0—according to socioeconomic characteristics of the invited job seekers elicited before treatment allocation. Figure A2 presents the distribution of predicted wages by treatment and gender of the job seekers who were invited to apply for the position. We find no significant differences in the predicted wages between the treatment and control groups (Wilcoxon rank-sum test: $p = .979$), indicating that randomization was also successful on predicted wages, unemployment, and opportunity cost.

Finally, to assess the causal effect of the AA treatment on the qualifications of the applicant pool, measured as expected wage $\hat{w}$, we estimate the following linear probability model separately for male and female job seekers:

¹⁴ Acosta, Muller, and Sarzosa (2015) use this data set to establish a link between personality traits and wages and labor market participation. They find that openness and reading proficiency are linked to hourly wages, while conscientiousness is strongly related to being employed. The latter we are able to replicate, as we show in table A4.

$$\begin{aligned} \text{completed}_i = & \alpha + \beta_1 \text{AA}_i + \beta_2 \hat{w}_i + \beta_3 \hat{w}_i^2 + \beta_4 \text{AA}_i \\ & \times \hat{w}_i + \beta_5 \text{AA}_i \times \hat{w}_i^2 + \varepsilon_i. \end{aligned} \quad (5)$$

The estimated coefficients of equation (5) are presented in table 6. In the control group, we find that application rates are higher for more-qualified men in two of the three experiments. No such pattern is detected for female job seekers. The introduction of affirmative action does not affect the qualifications level of male applicants in any of the three experiments. Yet we find evidence that affirmative action could deter qualified female job seekers from applying, significantly so in the A2 experiment.

RESULT 3: There is no detectable negative effect of affirmative action on the qualification level of male applicants in any of the experiments. There is some evidence of negative effects on qualified females.

D. Top Applicants by Treatment and Gender

When hiring, firms are often primarily interested not in the average qualifications of the applicant pool but in whether they are able to attract the best employees in their respective field. To address this question, we compare the predicted wage that a candidate in the 85th percentile of the distribution would receive under both treatments. Since discrimination could favor one gender over the other and result in wage gaps, we consider only the predicted wage that any candidate would receive had they been male. We report the differences in predicted log male wage in table 7. We do not find evidence that affirmative action significantly reduces the quality of the best applicants; in experiment A2, we even observe that the 85th percentile of applicants in the AA treatment is, on average, better qualified than that under the control (NOAA) treatment.

E. Individual Wage Disparities and Application Rates

To analyze the effect of affirmative action, it is important to consider how it affects sorting by job seekers who confront different degrees of discrimination. To address this question, we build an individual measure of wage disparity, D_i . On the basis of the estimated coefficients of equation (3), we predict the counterfactual wage that a job seeker would received—*ceteris paribus*—had she or he been of the opposite sex. The larger the earning gender gap—the difference between predicted wages for a male (female) job seeker had he (she) been female (male)—the larger the degree of wage disparity this person experiences in the labor market. To analyze how affirmative action affects the sorting of candidates who confront different degrees of discrimination, we estimate the following linear probability model for male and female job seekers:

Table 6
Linear Probability Model of Completion Rates on Predicted Wage

	Pooled		Assistant 1 (A1)		Assistant 2 (A2)		Consultant (C)	
	Male	Female	Male	Female	Male	Female	Male	Female
Affirmative action (AA)	-.028 (.045)	.100*** (.036)	.001 (.044)	.058* (.034)	-.063 (.042)	.103*** (.037)	.158 (.169)	.074 (.098)
Predicted wage	.001 (.016)	-.035 (.022)	.010 (.026)	-.054 (.033)	.017 (.019)	.029 (.043)	.112*** (.041)	.018 (.045)
(Predicted wage) ²	.003 (.004)	-.007 (.006)	.004 (.008)	-.001 (.009)	.012*** (.003)	-.017 (.022)	.028*** (.009)	.006 (.009)
AA × predicted wage	-.001 (.023)	.035 (.032)	-.014 (.028)	.046 (.044)	.023 (.032)	-.145*** (.046)	-.082 (.059)	.095 (.066)
AA × (predicted wage) ²	-.005 (.006)	.002 (.008)	-.011 (.011)	.018 (.019)	-.025 (.021)	-.075*** (.020)	-.011 (.015)	.014 (.013)
Constant	.463*** (.031)	.359*** (.025)	.443*** (.039)	.350*** (.034)	.548*** (.040)	.482*** (.069)	.009 (.118)	.180*** (.062)
Observations	842	893	323	398	335	324	157	136

NOTE.—This table reports the results of a linear model of the decision to complete the application process on centered log predicted hourly earnings. Standard errors are reported in parentheses. Significance levels are for the results of *t*-tests.

* $p < .1$.

*** $p < .01$.

Table 7
Difference in Expected Log Wage for the 85th Percentile

	Assistant 1 (A1)	Assistant 2 (A2)	Consultant (C)
NoAA – AA	.00716 (.0668)	–.117* (.0660)	.00179 (.00836)
Observations	109	109	48

NOTE.—A negative value of NoAA – AA means that the qualification level under affirmative action is higher than under no AA. Standard errors are reported in parentheses. Significance level is for the results of *t*-tests. AA = affirmative action treatment; NoAA = control treatment.

* $p < .1$.

$$\text{completed}_i = \alpha + \beta_1 \text{AA}_i + \beta_2 D_i + \beta_3 \text{AA}_i \times D_i + \varepsilon_i. \quad (6)$$

Table 8 presents the estimation results for equation (6). In the regression we control for the qualifications of the candidates, measured as predicted wage. We find that in the absence of affirmative action, application rates are significantly lower for male job seekers who have a larger advantage in the labor market in two of the three experiments. When there is no affirmative action, a 1% increase in wage discrimination results in a 1.1–2.3-percentage-point decrease in the application rate of male job seekers. The AA treatment changes the male sorting pattern significantly only in experiment A2, where we find that men apply less when confronted with affirmative action when they have a higher wage disparity in their favor. For female job seekers, we do not find any pronounced pattern of sorting associated with wage gaps, either in the absence or the presence of affirmative action.

VI. Discussion and Conclusion

Much has been written about how women are less likely to enter the labor force than men and how policies such as affirmative action might reduce this segregation. We have designed and conducted three field experiments to answer a very specific question. Does including affirmative action statements in the job description encourage women to apply for jobs and hence reduce the gender gap in applications? We find that it does, and the application pool under affirmative action is no less qualified than the application pool without affirmative action.

In the Colombian context of our field study, affirmative action is a voluntary choice for employers. In situations where affirmative action is compulsory, the sorting of applicants may be very different. For example, Seierstad and Opsahl (2011) provided evidence that the use of quota rules in the boards of publicly listed companies in Norway is associated with increased participation of women in multiple boards, reflecting short-term restrictions in the supply of female executives.

Although our experiments resolve important internal-validity concerns, compared to laboratory experiments, and may also offer local external validity in similar markets in developing and emerging economies, we cannot claim that these results hold globally in all labor markets. We see a necessity

Table 8
Completion Rates by Predicted Gender Wage Gap

	Pooled		Assistant 1 (A1)		Assistant 2 (A2)		Consultant (C)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Wage gender gap	-.027 (.042)	-.001 (.034)	-.230* (.133)	-.232 (.151)	-.012 (.033)	.009 (.057)	-.107* (.064)	-.044 (.048)
AA	-.026 (.045)	.103*** (.036)	-.009 (.070)	.024 (.080)	-.035 (.044)	.104** (.043)	.082 (.189)	.022 (.093)
AA × wage gender gap	-.019 (.057)	-.038 (.048)	.279 (.194)	.153 (.193)	-.139*** (.046)	-.009 (.069)	-.026 (.106)	-.095 (.071)
Predicted wage	.005 (.017)	-.035 (.022)	.035 (.031)	-.082* (.044)	.019 (.019)	.031 (.048)	.136*** (.039)	.032 (.044)
(Predicted wage) ²	.002 (.005)	-.007 (.006)	-.001 (.008)	-.033 (.028)	.012*** (.003)	-.016 (.027)	.024** (.009)	.007 (.008)
AA × predicted wage	.003 (.025)	.032 (.032)	-.043 (.044)	.063 (.068)	.069** (.029)	-.147*** (.048)	-.053 (.070)	.113* (.065)
AA × (predicted wage) ²	-.007 (.006)	-.001 (.009)	-.004 (.011)	.036 (.049)	-.039** (.018)	-.076*** (.021)	-.011 (.015)	.013 (.013)
Constant	.464*** (.031)	.360*** (.025)	.448*** (.048)	.404*** (.054)	.549*** (.040)	.480*** (.077)	-.026 (.110)	.149*** (.062)
Observations	842	893	323	398	335	324	157	136

NOTE.—This table reports the results of a linear model of predicted individual gender wage gap and affirmative action (AA) on completing the application process. Standard errors are reported in parentheses. Significance levels are for the results of *t*-tests.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

for similar experiments in labor markets with higher gender equity, where women may perceive affirmative action as disempowering, and markets with different levels of unemployment and unemployment differentials. Furthermore, it would also be interesting to determine whether states that do not support, or even actively hinder, female labor force participation would see similar results from voluntary affirmative action rules—if allowed.

Affirmative action statements in job advertisements may vary in their effectiveness at closing the gender gap in applications, depending on the context. For example, if there are cultural or childcare constraints that vary across economies, then affirmative action might be insufficient to induce women to join the labor market. It is therefore heartening to find such evidence in our study in Colombia. Further replications are necessary to test whether the effectiveness of affirmative action depends on region-specific factors and which of those factors are most important.

Appendix

A1. Job Advertisements

Assistant Experiment 1

Newspaper. A German university is looking for research assistants to work in rural areas. More information is available at: [http://\[. . .\]](http://[. . .]).

Employment Boards. The Centre for the Study of Poverty, Equity and Growth from University of Goettingen is looking for research assistants to work in rural areas in Colombia. It is expected that the assistants will live in rural communities for two to three months. Research assistants would help to connect rural communities and principal researchers. Some of the tasks required include:

- Establish contact with local institutions;
- Conduct interviews and surveys;
- Collect secondary data;
- Recruit, train and supervise local enumerators;
- Write field work reports.

We are looking for professionals from any area of studies who have recently graduated (or are about to complete the degree). The candidates should be motivated to work in rural areas, should have capacity to work semi-independently, be proactive, good team members and have good communication skills. Although experience in similar tasks is preferable it is not compulsory. If you are interested and want to know more about the position, please register at: [http://\[. . .\]](http://[. . .]).

Assistant Experiment 2

Newspaper. A German university is looking for research assistants to work in rural areas. For more information go to: [http://\[. . .\]](http://[. . .])

Employment Boards. The Centre for the Study of Poverty, Equity and Growth from University of Goettingen is looking for university research assistants from any area of studies. Research assistants would work in Bogotá supporting our research group in different activities that include:

- Establish contact with local institutions;
- Conduct interviews and surveys;
- Data entering;
- Collect secondary data.

No previous experience is required. If you are interested, please register at: [http://\[. . .\]](http://[. . .]).

Consultant Experiment

Newspaper. Consultant with at least 2 years of experience is required to work in the implementation, follow-up, and evaluation of municipal development plans using participatory methods.

E-mail. Urgent! Consultant firm with ample experience in public and private sector is looking for a Consultant to work in implementation, follow-up, and evaluation of municipal development plans using participatory methods. At least two years of experience is required. More information at: [http://\[. . .\]](http://[. . .]).

A2. Socioeconomic Information

Table A1
Variables Collected at the Statement of Interest

Variable	Assistant 1	Assistant 2	Consultant
Personal information:			
National ID number (Cedula)		X	X
Gender	X	X	X
Marital status	X	X	X
Do you have a master's degree?	X	X	X
How many years of experience do you have?			X
Day of birth	X	X	
Place of birth	X		
Actual address	X		
Permanent residence	X		
Time availability for next year	X		
Highest education level:			
Institution	X	X	
University	X	X	
Area of studies	X	X	
Titles	X	X	
Years of graduation	X	X	
Average grades		X	
Family information:			
Father's name (first name, last name)	X		

Table A1 (Continued)

Variable	Assistant 1	Assistant 2	Consultant
Address (street, barrio, city, municipality, department)	X		
Mothers's name (first name, last name)	X		
Address (street, barrio, city, municipality, department)	X		
Children		X	
Academic history:			
University/college	X	X	
City	X		
From/to (dates)	X		
Degree	X	X	
Year	X	X	
Health information:			
Do you have medical insurance? (yes/no)	X		
Is your medical insurance valid for outside Bogotá? (yes/no)	X		
Vaccinations: tuberculosis, tetanus, diphtheria, yellow fever, hepatitis B	X		
Where did you find the job offer? (e-mail, poster, web portal, newspaper, other)	X		
Personality test (Big 5)		X	
Risk aversion: you like to take risks (10) or you avoid them (1)		X	
Impulsiveness: you think a lot before you take a decision (1) or you are impulsive (10)		X	
Time preferences: you are impatient (1) or patient (10)		X	
Cognitive-reflection test		X	
Work-related ability		X	

A3. Invitation Letters (Translated from Spanish)

A3.1. Assistant 1

Thank you for your interest in working in our research group. Due to the high number of applications the screening process took a couple of days longer than we had anticipated. We apologize for this delay.

The position is available for a research assistant to work in rural areas in Santander with tobacco farmers. You will work in a team where their duties include: establish contacts with local institutions; conduct interviews and surveys; collect secondary data; recruit, train, and supervise local enumerators; and write reports on field activities.

The contract is for two months with possibility of extension, depending on the duration of the project and the candidate's performance. The base salary is 1,500,000 pesos plus travel commissions of 65,000 for each day in the field. It is expected that the research assistant will be in the field for two months. Additional costs, such as transportation, will also be covered by the project.

If you are interested in applying for this position, please complete the form below. It takes approximately 60 minutes to answer it. At any time you can stop and resume the process. When you resume, you can continue without losing information as long as you have saved the changes. Please be completely honest in answering the following questions. This will allow us to determine your compatibility for work in our research group.

AFFIRMATIVE ACTION STATEMENT

The University of [hidden] is committed to the policy of equal opportunities in the search and selection of staff. We seek to increase women's participation in areas where so far they have been underrepresented. At least 50% of the research assistants hired will be women.

To apply for the job, please click the following [link].

The deadline to complete the application is December 31, 2010. We will contact you soon indicating whether you have been selected for interview. For additional questions please contact us at [e-mail].

Best regards, [Name].

A3.2. Assistant 2

Thank you for your interest in working in our research group. Due to the high number of applications we would need to do a second round of tests. All the tests will be carried out in this internet platform. If you are interested in the position please complete the form. Completing the form take about 1 hour and includes different tests on work-related abilities. Please complete the test individually and without asking for support to other people. Do not use calculators or other electronic devices.

You can interrupt the application process in any moment. In order not to lose the information, please save the changes. Please be completely honest while answering the questions. This would allow us to determine your compatibility with the job offer. The deadline to complete the application form is August 13th. The best candidates will be invited for interview.

The research assistants would be hired to enter data from paper-based surveys. The payment will be by survey entered. The data entered would be verified. Hence, your payment would depend on your job performance.

We have flexibility to adjust the working hours such that students do not have problems with classes. Nonetheless, all the work needs to be done in our offices. As this is a contract by completed activities, there is no specified duration of the contract specified. We estimate that completing the data entering will take between 1 and 2 months. The work will start in the next two weeks.

AFFIRMATIVE ACTION STATEMENT

The University [hidden] uses an equal opportunity policy in the recruitment and employment of the personal. We aim at increasing female representation in areas where they have been underrepresented. At least 50% of the positions will be filled by women.

If you want to apply, enter [here].

If you do not want to continue, please press [here].

If you have additional questions, please do not hesitate in contacting us.

A3.3. *Consultant*

[The Company] Ltd. is a consulting firm with more than 15 years of experience working for the public and private sectors. We seek professionals with at least 2 years of experience to work as consultants under service provision contracts on a consulting project.

The successful candidate will work on the implementation, participatory monitoring, and evaluation of municipal development plans in two municipalities of Santander. The contract is for 4 months. The monthly salary is 3 million pesos, negotiable. Besides the needed experience, it is indispensable that applicants are willing to travel outside of Bogotá. We seek professionals preferably in public administration, international relations, and economics.

AFFIRMATIVE ACTION [presented only to affirmative action group]:

[The company] is committed to the policy of equal opportunities in the search and selection of staff. Thus, [the company] seeks to increase women's participation in areas where so far they have been underrepresented. In case of candidates with equal qualification level, women will be preferred.

If you are interested in applying for this position, please complete the form below. It takes approximately 40 minutes to answer the form. At any time you can interrupt the process. When you resume, you can continue without losing information, as long as you have saved the changes. The deadline to complete the application is January 21, 2011. We will contact individuals who have been selected for interview very soon. For any additional questions please contact us at [e-mail].

A4. Tests Included in Application Process

A4.1. *Frederick Cognitive-Reflection Test*

Would you please answer the following questions.

A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost? _____ cents.

If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? _____ minutes.

In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake? _____ days.

A4.2. *Work Example*

Subjects were asked to digitize the example questionnaire presented in figure A1.

Cuestionario
Por favor, digite la información contenida en la encuesta en los campos correspondientes

Descriptivo encuesta			
1.1. Formulario No.	BO-13	1.2. Resultado	L+V+P
1.4. Fecha	08-02-11	1.5. Municipio	CART
1.7. Encuestador	Belky Velasco	1.6. Vereda	ALDO SANCHEZ

1.1 Formulario No

1.2 Resultado

1.3 Hora de Inicio

1.4 Fecha

1.5 Municipio

1.6 Vereda

1.7 Encuestador

1.8. Productor

1.9. Cédula de ciudadanía

1.10. Teléfono

1.11. Encuestado

1.12. Género ☐ Mujer ☒ Hombre

1.13. Cultiva tabaco ☒ Si ☐ No

1.14. Relación con jefe de hogar?

1.8 Productor

1.9 Cédula de ciudadanía

1.10 Teléfono

1.11 Encuestado

1.12 Género

1.13 Cultiva tabaco

1.14 Relación con jefe de hogar

3. ¿Cuántos años tienen el padre y la madre del niño?

a. Padre: años b. Madre: años

4. ¿Cuántos años de colegio/escuela aprobaron los padres?

a. Padre: años b. Madre: años

	Padre	Madre
3. Cuántos años tienen		
4. Cuántos años de colegio		

5. ¿Hablando en general, en una escala de uno a seis, cuanto confianza tiene en cada uno de los siguientes grupos? (1=usted tiene confianza total y 6=usted no tiene confianza en absoluto)

Grupo	Escala de confianza (de 1 a 6)
La gente en general	3
Sus vecinos	3
El gobierno local	2
La policía	2
Los medios de información (periódicos, noticieros, etc)	1
El gobierno nacional cumpliendo sus promesas	0
El gobierno nacional ayudando a los pobres	1

En estos campos sólo pueden introducirse números

La gente en general

Sus vecinos

El gobierno local

La policía

Los medios de información

El gobierno nacional cumpliendo sus promesas

El gobierno nacional ayudando a los pobres

FIG. A1.—Work example screen shots.

A5. Metropolitan Areas

Bogotá area: Cajicá, Chía, Cota, Facatativá, Soacha.

Medellín area: Barbosa/Antioquia, Bello, Copacabana, Envigado, Girardota, Itagüí, La Estrella, Sabaneta.

Caribbean area: Barranquilla, Candelaria, Ponedera, Cartagena, Santa Marta, Montería, Cereté, Planeta Rica, San Carlos/Córdoba.

Cali area: Cal, Palmira, Yumbo, Jamundí.

Coffee area: Chinchiná, Villamaría, Manizales, Dosquebradas, Santa Rosa de Cabal, Calarcá, Filandia, Pereira.

A6. Representativeness of the Sample

We compare socioeconomic characteristics of the job seekers in our experiment with those in the 2010 RGUS survey from the Colombian Ministry of Education. For the A1 and A2 experiments, we take as reference university graduates from any area of study over the last year, whereas for the C experiment, we compare with graduates with 3–5 years of experience from areas of studies relevant for the job. Table A2 presents the results.

We find that the proportion of female job seekers in two of the three experiments (A2 and C) is not comparable with that in RGUS survey. This could be related to the characteristics of the job. For example, A2 explicitly required computer skills, a task that is considered more male oriented. The C experiment required availability to travel outside the capital and spending a considerable amount of time in the field, which could have restricted women from applying.

We also find significant differences with respect to degree and area of studies. This is not very surprising, as the announcement in the A1 and A2 experiments emphasized that we were hiring recently graduated students and the description of the job required very basic skills. In the C experiment, the share of job seekers with a master's title is relatively higher than that in the RGUS survey, which can be explained by the requirement of a more qualified subject pool. Regarding area of studies, we find that participants in the A1 and A2 experiments were significantly more likely to have a background in agricultural sciences and social sciences and were less likely to be economists or engineers, compared with the RGUS sample. This is also explained by the type of job that we announced, which required work in rural areas in Colombia conducting interviews.

Table A2
Descriptive Statistics of Job Seekers

	Assistant 1 (A1)	Assistant 2 (A2)	Consultant (C)	Survey
A. Gender:				
Male	.4447	.4954	.5358	.4353
Female	.5553	.5046	.4642	.5647
χ^2		22.452 (<.001)		
B. Academic degree:				
Bachelor	.9018	.9553	.6645	.8596
Master	.0982	.0447	.3355	.1404
χ^2		167.362 (<.001)		
C. Study area:				
Agricultural science	.0962	.0559		.0137
Fine arts	.0151	.0218		.0459
Education science	.0165	.0287		.0556
Health sciences	.0165	.0505		.0656
Social sciences	.4217	.3765		.1750
Economics and business	.1511	.1242	1.0000	.3105
Engineering	.2184	.1201		.3113
Natural sciences	.0646	.1255		.0226
Other	.0000	.0969		.0000
χ^2		3,856.426 (<.001)		

NOTE.—This table reports the composition of the job seekers in the first stage between the A1, A2, and C experiments and the Survey of Recently Graduated University Students in Colombia conducted by the Ministry of Education in 2010 and available online (<http://www.graduadoscolombia.edu/co>). The relevant area of study for the C experiment was narrowly defined; therefore, all candidates are from economics, business, and administration studies. The χ^2 -test compares the distribution in the experiment with that in the survey of recently graduated university students

A7. Predicted Wage

Table A3
Variables in Assistant 2 Experiment versus Skills Toward Employability and Productivity (STEP) Survey

Variable	Experiment	STEP Survey (World Bank 2014)	
		Explanation	Transformation
Wage		Last wage earned	Log
Experience	Years of work after graduation	Years of work after graduation	
Risk aversion	GSOEP question	Similar to GSOEP question	Standardized: mean 0, variance 1
Age	At time of filling in application	At time of filling in survey	
Personality	Big 5, Spanish version	Big 5, Spanish version	Standardized: mean 0, variance 1
Gender	Male/female		

NOTE.—GSOEP = German Socioeconomic Panel.

Table A4
Heckman Two-Stage Wage Equations Using Skills Toward Employability and Productivity (STEP) Survey Data

	Model for A1			Model for A2			Model for C		
	Wage	Selection	Mills	Wage	Selection	Mills	Wage	Selection	Mills
Higher education	.206* (.117)	-.009 (.129)		.170* (.100)	.037 (.135)		.221 (.148)	-.020 (.126)	
Graduate	.426*** (.127)	.062 (.144)		.394*** (.106)	.032 (.151)		.394** (.168)	.131 (.139)	
Postgraduate	1.371*** (.311)	.182 (.413)		1.315*** (.255)	.247 (.427)		1.248*** (.420)	.409 (.416)	
Experience	-.030 (.086)	.313** (.124)		.021 (.052)	.293** (.137)		-.054 (.112)	.309** (.120)	
Age	-.006 (.007)	.020*** (.004)		-.000 (.004)	.020*** (.005)				
Age ²	-.000 (.000)	.000*** (.000)		-.000 (.000)	.000*** (.000)				
Female	.107 (.252)	-.214 (.197)		.054 (.207)	-.147 (.218)		.246 (.347)	-.572*** (.094)	
Higher education × female	-.169 (.184)	.217 (.160)		-.069 (.141)	.179 (.166)		-.210 (.238)	.216 (.157)	
Graduate × female	.042 (.201)	.185 (.190)		.167 (.162)	.208 (.196)		.028 (.244)	.120 (.185)	
Postgraduate × female	-.457 (.428)	.472 (.487)		-.196 (.331)	.373 (.500)		-.491 (.529)	.322 (.489)	
Female × experience	.036 (.080)	-.142 (.140)		-.001 (.058)	-.128 (.152)		.056 (.104)	-.153 (.137)	
Female × age	.002 (.006)	-.012** (.005)		-.002 (.005)	-.011* (.006)				
Female × age ²	.000 (.000)	-.000* (.000)		-.000 (.000)	-.000 (.000)				

Table A4 (Continued)

	Model for A1			Model for A2			Model for C		
	Wage	Selection	Mills	Wage	Selection	Mills	Wage	Selection	Mills
Big 5:									
Conscientiousness				-.054 (.046)	.138*** (.053)				
Extraversion				-.036 (.045)	.087 (.058)				
Agreeableness				.007 (.043)	-.056 (.061)				
Emotional stability				-.044 (.043)	.043 (.056)				
Openness				.102** (.042)	.041 (.057)				
Risk aversion				.095** (.040)	-.027 (.053)				
Female × conscientiousness				-.028 (.057)	-.068 (.069)				
Female × extraversion				.102* (.061)	-.127* (.072)				
Female × agreeableness				-.014 (.060)	-.007 (.075)				

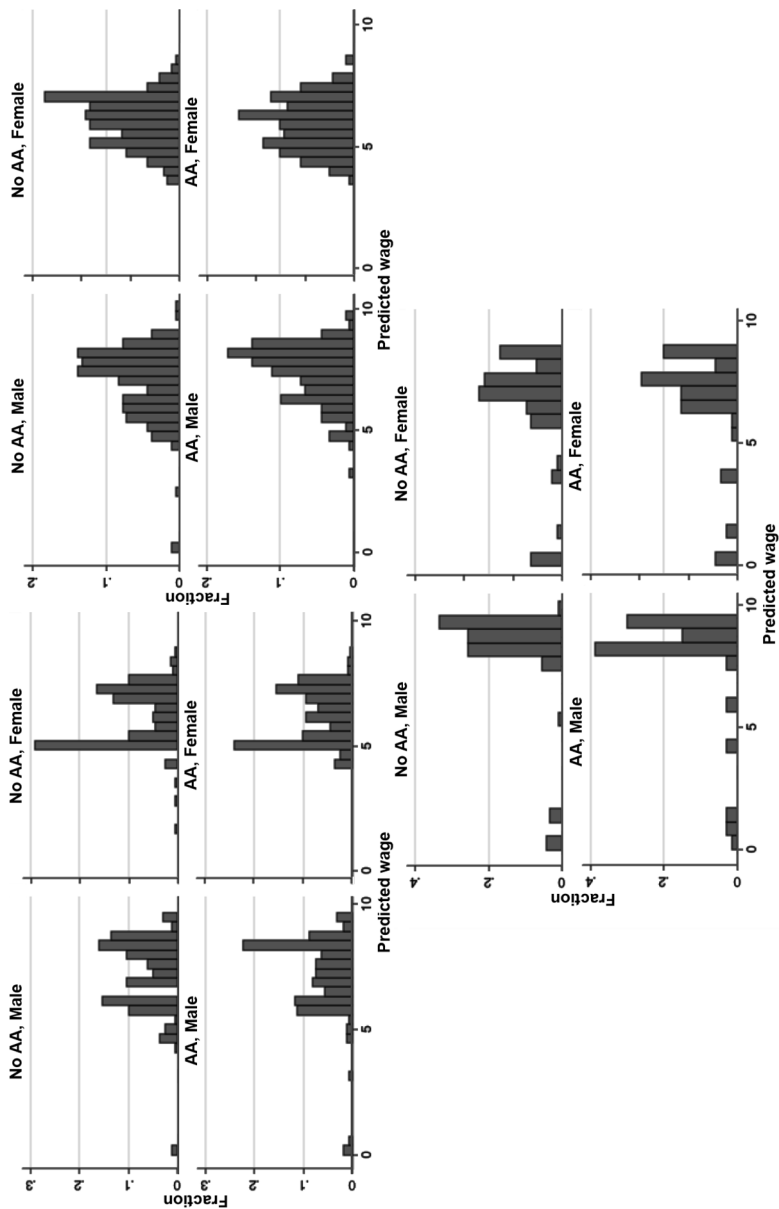


FIG. A2.—Distribution of predicted wages from the Skills Toward Employability and Productivity (STEP) survey. The differences between the treatments by gender are not significant (Wilcoxon rank-sum tests: male, $p = .477$; female, $p = .480$). AA = affirmative action treatment.

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